

Chapter 11 Assignment — Conic Sections

Name: _____

Date: _____

How to use this set: Each topic begins with *Strategy Notes* and a *Worked Example*.

Then try the *Practice* (skill) and the *Challenge* (deeper) items. When an equation is requested, give the standard form and clearly label key features (center/vertex, foci, axes, asymptotes, directrix).

1. Topic 1: Parabolas

Strategy Notes. Standard forms: $(x-h)^2 = 4p(y-k)$ (opens up/down) and $(y-k)^2 = 4p(x-h)$ (opens right/left). The vertex is (h, k) , the focus is $(h, k+p)$ or $(h+p, k)$, and the directrix is $y = k-p$ or $x = h-p$ respectively. The parameter p is the directed distance from vertex to focus.

Worked Example. For $y^2 = 8x$, $4p = 8 \Rightarrow p = 2$. Vertex $(0, 0)$, focus $(2, 0)$, directrix $x = -2$, axis $y = 0$.

Practice.

- (a) Find the focus, directrix, and axis of symmetry of $x^2 = 8y$. Sketch the graph and mark the latus rectum endpoints.
- (b) Find an equation of the parabola with vertex $(1, -2)$ and focus $(1, 1)$. State its axis and directrix.

Challenge.

- (a) Find the equation of the parabola with focus $(3, 2)$ and directrix $y = -1$. Write your answer in the form $(x - h)^2 = 4p(y - k)$.

- (b) Convert $y = -x^2 + 6x - 10$ to standard form and determine its vertex, focus, and directrix.

2. Topic 2: Ellipses

Strategy Notes. In standard position and centered at (h, k) : $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$ with $a \geq b > 0$. Vertices are $(h \pm a, k)$ or $(h, k \pm a)$. Foci satisfy $c^2 = a^2 - b^2$; they lie on the major axis.

Worked Example. For $\frac{x^2}{9} + \frac{y^2}{4} = 1$, $a = 3$, $b = 2$, so $c = \sqrt{5}$. Vertices $(\pm 3, 0)$, co-vertices $(0, \pm 2)$, foci $(\pm \sqrt{5}, 0)$.

Practice.

- (a) For $\frac{x^2}{25} + \frac{y^2}{9} = 1$, find the vertices, co-vertices, and foci. State the lengths of the major and minor axes.

- (b) Find an equation for the ellipse with foci $(\pm 5, 0)$ and minor axis length 6.

Challenge.

- (a) An ellipse has center $(2, -1)$ and vertices $(2 \pm 7, -1)$. Its minor axis has length 10. Find the equation and the foci.
- (b) The sum of distances to the foci is 10. The ellipse is centered at $(0, 2)$ with a *vertical* major axis and passes through $(3, 2)$. Find its equation.

3. Topic 3: Hyperbolas

Strategy Notes. Centered at (h, k) , horizontal transverse axis: $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$
with asymptotes $y - k = \pm \frac{b}{a}(x - h)$. Vertical transverse axis: $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$
with asymptotes $y - k = \pm \frac{a}{b}(x - h)$. Foci satisfy $c^2 = a^2 + b^2$.

Worked Example. For $\frac{x^2}{9} - \frac{y^2}{4} = 1$, $a = 3$, $b = 2$, so $c = \sqrt{13}$. Vertices $(\pm 3, 0)$; asymptotes $y = \pm \frac{2}{3}x$; foci $(\pm \sqrt{13}, 0)$.

Practice.

(a) For $\frac{y^2}{16} - \frac{x^2}{9} = 1$, find the vertices, foci, and equations of the asymptotes.

(b) Find the equation of the hyperbola with a horizontal transverse axis, vertices $(\pm 5, 0)$, and asymptotes $y = \pm 2x$.

Challenge.

- (a) A hyperbola centered at the origin has one focus at $(0, 13)$ and one vertex at $(0, 5)$. Find its equation and asymptotes.

- (b) By rotating axes by 45° , put $xy = 4$ into standard form. State the type of conic and its eccentricity.

4. Topic 4: Shifted Conics

Strategy Notes. For $Ax^2 + Cy^2 + Dx + Ey + F = 0$ with $B = 0$ (no xy term), group x 's and y 's and complete the square to identify the type and parameters. Signs of A and C distinguish ellipse/circle ($A, C > 0$), hyperbola ($AC < 0$), or parabola (exactly one of A, C nonzero).

Worked Example. $9x^2 + 4y^2 - 36x + 24y + 36 = 0 \Rightarrow 9(x^2 - 4x) + 4(y^2 + 6y) + 36 = 0$.
Completing the square gives $\frac{(x-2)^2}{4} + \frac{(y+3)^2}{9} = 1$ (ellipse centered at $(2, -3)$).

Practice.

- (a) Classify and convert to standard form: $x^2 + y^2 - 6x + 8y - 11 = 0$. State the center and radius/axes lengths.
- (b) Write $4x^2 - 9y^2 + 16x - 18y - 11 = 0$ in standard form and list its center, vertices, and asymptotes.

Challenge.

- (a) Classify and put in standard form: $16x^2 + 9y^2 - 64x + 54y + 1 = 0$. Identify the lengths of the major and minor axes and the foci.
- (b) Determine the value(s) of k for which $x^2 + kxy + y^2 - 2x - 8y + 1 = 0$ represents (i) an ellipse, (ii) a parabola, or (iii) a hyperbola. Justify using the discriminant $\Delta = B^2 - 4AC$.

5. Topic 5: Rotation of Axes

Strategy Notes. For a general quadratic $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ with $B \neq 0$, choose θ so that $\tan(2\theta) = \frac{B}{A - C}$. Then $\begin{cases} x = u \cos \theta - v \sin \theta, \\ y = u \sin \theta + v \cos \theta \end{cases}$ eliminates the uv -term. Translate if needed to complete the square.

Worked Example. Remove the xy -term from $3x^2 - 4xy + 3y^2 = 36$. Since $\tan(2\theta) = \frac{-4}{0}$, take $\theta = 45^\circ$. With $x = \frac{u-v}{\sqrt{2}}$, $y = \frac{u+v}{\sqrt{2}}$, the equation becomes $u^2 + 5v^2 = 36$ (an ellipse in the rotated coordinates).

Practice.

- (a) Diagonalize $5x^2 + 12xy + 5y^2 = 100$ by a 45° rotation and identify the conic in the (u, v) -plane.

- (b) Find the rotation angle that eliminates the xy -term in $2x^2 - 4xy + 5y^2 = 1$, then state whether the resulting conic (in rotated coordinates) is an ellipse, hyperbola, or parabola.

Challenge.

- (a) Show that the discriminant $\Delta = B^2 - 4AC$ is invariant under rotation and use it to justify that $\Delta = 0$ corresponds to a parabola (after an appropriate translation).

6. Topic 6: Polar Equations of Conics

Strategy Notes. With focus at the origin and directrix a vertical line $x = \pm d$ or horizontal line $y = \pm d$, the polar equation is

$$r = \frac{ed}{1 \pm e \cos \theta} \quad (\text{vertical directrix}), \quad r = \frac{ed}{1 \pm e \sin \theta} \quad (\text{horizontal directrix}).$$

Here e is the eccentricity: $0 < e < 1$ (ellipse), $e = 1$ (parabola), $e > 1$ (hyperbola).

Worked Example. For $e = \frac{1}{2}$ and directrix $x = -3$, $r = \frac{(1/2) \cdot 3}{1 - (1/2) \cos \theta} = \frac{3}{2 - \cos \theta}$.

Practice.

- (a) Find a polar equation for the conic with focus at the origin, eccentricity $e = \frac{3}{2}$, and directrix $x = 4$.

- (b) Identify the type of conic and its directrix for $r = \frac{5}{1 - 2 \cos \theta}$.

Challenge.

- (a) Determine e and the directrix for $r = \frac{4}{2 + \sin \theta}$. Then classify the conic.
- (b) Decide whether $r = \frac{3}{1 + 0.6 \cos \theta}$ is an ellipse, parabola, or hyperbola. Find e and a corresponding directrix.

7. Focus on Modeling: Conics in Design

- (a) A satellite dish is a parabolic reflector 8 m across at the rim and 1 m deep at the center. Place the vertex at the origin with the dish opening upward. Find the focal length p and determine where to place the receiver (the focus) along the axis. *Hint:* If the rim is at $y = 1$ with radius 4, then $x^2 = 4py$ at $y = 1$.
- (b) An archway is shaped like the top half of an ellipse centered at the origin. It is 12 m wide at the base and 4.5 m high at the center. A sign of height 3 m is to be hung vertically from the center. How wide is the opening at the height of the bottom of the sign? Give an exact expression and a decimal approximation.